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Recovery Subsystem

Log-Based Recovery

A log is a sequence of log records.

What does log stand for?

Log is a history of all changes made to DB. Whatever actions you take in a program, it is kept as a log. If any error occurs in a program that you are unable to understand, even the compiler cannot decipher it then we check the log record and we can understand where the error/problem has arised. Log helps us in debugging.

The records keep information about update activities on the database. (insert/update/delete)

The log is kept on stable storage (previous stable state - Roll back)

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// Log record starts from <Ti start>
i stands for the no. of transaction.
i = 0, 1, 2, 3 ....

$\langle T_i, X, V1, V2 \rangle$

New value

Transaction Variable Old value

V1 will be updated by V2, if we cannot update it then it will rollback.

If the update is successful then
<Ti commit>

We go from one stable state to another with the help of commit.

If we cannot go to committed state then what are the recovery mechanisms.

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Types of transaction :

(i) undo / rollback (Ti) log

(ii) redo (Ti) log - If the value of a particular transaction is not updated, then we try the transaction again updated by Ti to the new values going forward from the first log record for Ti.

→ restores the value of all data items updated by Ti to their old values going backwards from the last log record for Ti.

Each item a data item X is restored to its old value V a special log record <Ti, X, V> is written out.

When undo of a transaction is complete a log record <Ti, abort> is written out.

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Undo - go back to old values

<T1, A, 1000, 950>

Transaction fails, A = 1000

<Ti, X, V> - records rollback action.

Redo means reapply the changes.

<T1, A, 1000, 950>

A = 950

UNDO - transaction didn't commit.

REDO - transaction committed but changes not saved to disk.